Examiner only

Option A: Immunology and Disease

7. Lyme disease, or Lyme borreliosis, is caused by the bacteria *Borrelia burgdorferi* and is endemic in some areas. There are a number of strains of this bacterium. The bacteria are transmitted to humans through the bite of an infected tick. Reported cases in England and Wales rose from 268 in 2001 to 959 in 2011, with an increase in the number of cases in Snowdonia National Park.

Ticks are tiny, spider-like creatures found in woodland and heath areas. They feed on the blood of birds and mammals, including humans. Ticks that carry the bacteria responsible for Lyme disease are found throughout the UK and in other parts of Europe and North America.

(a) State what is meant by the term *endemic*. [1]

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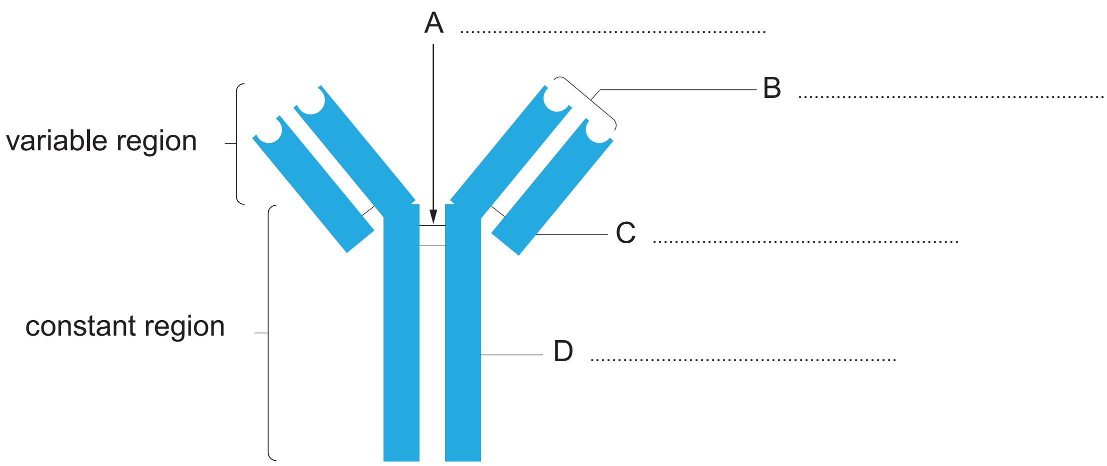
The immune system of an infected person will react by producing specific antibodies to the bacterial antigens.

(b) (i) Label the diagram below to identify the regions of the antibody. [3]



(ii) Define the term *antigen*, and explain the meaning of an antigen-antibody complex. [2]

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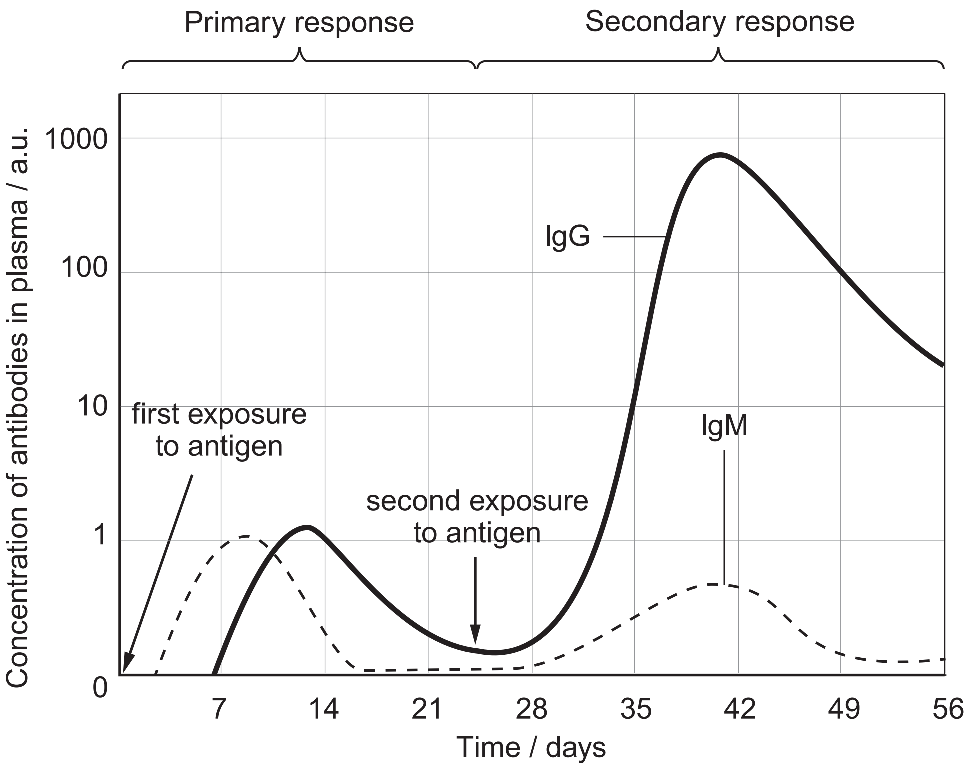
(c) The body's response to infection shows both a primary response and a secondary response to the foreign antigens. Both responses can be identified by measuring the levels of two antibodies found in the blood, IgM and IgG.
IgM is produced mainly by B-cells on initial exposure; IgG antibodies are produced in higher levels during the secondary response.



(i) Explain why the level of IgG is higher after the second exposure to the antigen than the first exposure. [1]

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(ii) Lyme disease can persist in patients for many years and the IgM levels remain high. Suggest possible reasons why. [2]

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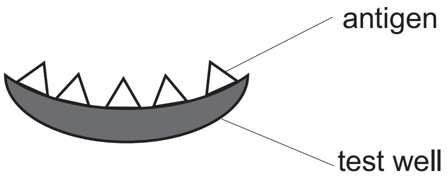
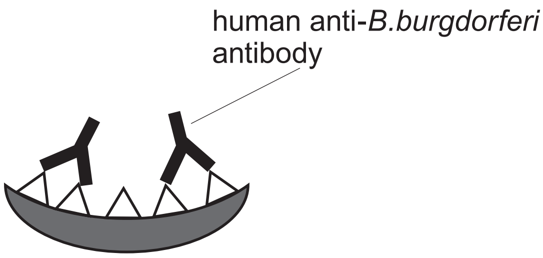
(d) The symptoms of Lyme disease include fever, chills, fatigue, headaches and muscle aches. As a consequence, patients with Lyme disease are often misdiagnosed.

An ELISA (enzyme-linked immunosorbent assay) can be carried out to diagnose Lyme disease using a blood sample from the patient around **two weeks** after infection. It is based on detecting the antibodies made in response to being exposed to *B.burgdorferi*.

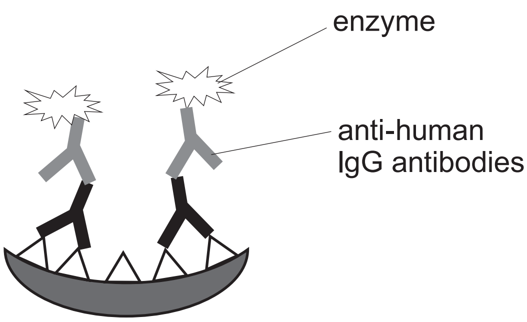
The results of the assay show the concentration of IgG antibodies in the patient's blood. The steps involved in the ELISA are shown below.



Specific antigen from *B.burgdorferi* attached to the surface of the test well.



Patient's blood plasma added to the well. Human antibodies bind to the *B.burgdorferi* antigen.



Enzymes attached to anti-human IgG antibodies cause a colour change if an antigen-antibody complex forms. Greater colour intensity indicates a higher IgG concentration.

(i) Using the graph, explain why carrying out an ELISA would not detect the IgG antibodies in the days immediately after the tick bite. [2]

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(ii) The ELISA can produce false negative results for Lyme disease. In a false negative result there is no colour change, even if the patient is infected with the bacteria.

Suggest why, even if there are anti-*B.burgdorferi* antibodies in the patient's plasma, the enzyme may not cause a colour change. [5]

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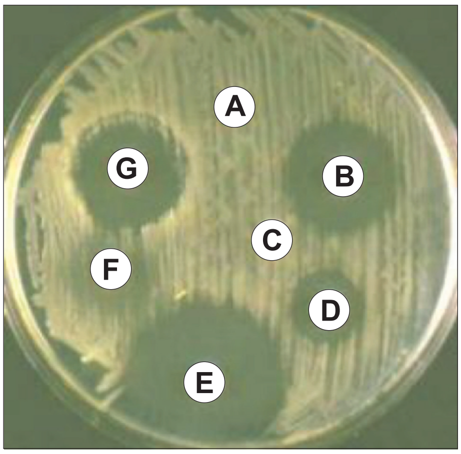
(e) *B.burgdorferi* is a Gram-negative bacterium and can be treated using a bacteriostatic antibiotic which stops protein synthesis.
The image below is of an agar plate showing the results of testing various antibiotics on *B.burgdorferi*.



(i) State the temperature at which the culture should be incubated.
Explain your answer. [1]

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(ii) The most effective antibiotic was found to be E. The diameter of the zone of inhibition was 22 mm. Calculate the area of the zone of inhibition caused by E to a suitable level of precision. [2]

Formula for the area of a circle: πr^2

$\pi = 3.14$

answer = ...

(iii) With reference to the image, suggest what assumption is being made when making this calculation. [1]

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